Imaginary-Axis Coefficient Type and the Origin (Exercise XXXII)

Anonymous Author(s)

Abstract

We compute the origin of the split-complex Euler circle frame with the coefficient type of the imaginary axis made explicit. The imaginary axis represents the imaginary part of the complex numbers; its coefficients may be taken as natural numbers (discrete axis $\{n \cdot i : n \in \mathbb{N}\}$) or as real numbers (continuous axis $\{t \cdot i : t \in \mathbb{R}\}$). Four machine-checked theorems: under natural coefficients, the real axis meets the imaginary axis at $\{(0,0)\}$ (solving $x = n \cdot i$ forces $x = 0$ and $n = 0$); under real coefficients, the intersection is again $\{(0,0)\}$ (solving $x = t \cdot i$ forces $x = 0$ and $t = 0$); the transposition maps the real axis onto the imaginary axis — the transposed real axis is the imaginary axis with real (continuous) coefficients; and the transposition preserves natural coefficients (the natural imaginary axis maps to the natural real axis). The origin is $(0,0)$ under both coefficient conventions. Zero errors, zero `sorry`.

Keywords: Lean, formalization, split-complex, coefficient type, origin, transposition, 0pat

1 The computation

The orthogonal Euler circle frame (Exercises XXVII–XXXI) lives in the split-complex plane $a + bj$, $j^2 = +1$. The imaginary axis represents the imaginary part of the complex numbers. This exercise makes the coefficient type of the imaginary axis explicit and computes the origin under each convention.

2 The two computations

Under natural coefficients (origin_natural_coefficients). The imaginary axis is the discrete set $\{n \cdot i : n \in \mathbb{N}\}$; the real axis is the continuous real line. A point on both axes has the form x (real axis) and $n \cdot i$ (natural imaginary axis); solving $x = n \cdot i$ forces $x = 0$ and $n = 0$. The intersection is the single point $(0,0)$. Under natural coefficients, the origin is $(0,0)$.

Under imaginary-part (real) coefficients (origin_imaginary_part). The imaginary axis is the continuous set $\{t \cdot i : t \in \mathbb{R}\}$. Solving $x = t \cdot i$ forces $x = 0$ and $t = 0$. The intersection is again the single point $(0,0)$. Under real coefficients, the origin is $(0,0)$.

The two computations agree: the origin of the frame is $(0,0)$ whether the imaginary axis is discrete (natural coefficients) or continuous (real coefficients). The natural coefficient $n = 0$ and the real coefficient $t = 0$ are the unique solutions of their equations.

3 The transposition

The real axis transposes to the imaginary axis (transpose_realAxis_eq_imaginaryAxis). The transposition $(a,b) \mapsto (b,a)$ maps the real axis $\{(x,0)\}$ onto the imaginary axis $\{(0,x)\}$: the transposed real axis is the imaginary axis, with real (continuous) coefficients. If the coefficients of the imaginary axis are locked to natural numbers, the transposed real axis is the imaginary axis computed by imaginary part (continuous) — the transposition itself does not discretize.

Natural coefficients are preserved by transposition (transpose_natural_imaginary_is_natural_real). The natural imaginary axis $\{n \cdot i\}$ maps under transposition to the natural real axis $\{(n,0)\}$. Discrete stays discrete: transposition changes the axis, not the coefficient type.

4 Honest boundary and position

The content is elementary algebra; the value is the explicit separation of the two coefficient conventions and the proof that they agree on the origin. The result concerns the frame; no claim is made about the Riemann zeta function. The analysis of ζ remains the open span of the bridge.

5 Formalization

The proof is a single Lean file (`proof.lean`, 175 lines), importing only `Mathlib.Data.Real.Basic` and basic tactics. The split-complex structure (30 lines, as in Exercises XXVII–XXXI) is self-contained. Four theorems, compiled with Lean 4.32.2 / mathlib v4.32.2, zero errors, zero `sorry`.

Theorem	Statement
`origin_natural_coefficients`	axes meet at $\{(0,0)\}$ ($n \in$
`origin_imaginary_part`	axes meet at $\{(0,0)\}$ ($t \in$
`transpose_realAxis_eq_imaginaryAxis`	T(real axis) = imaginary
`transpose_natural_...`	natural coefficients preser

6 Conclusion

With the coefficient type of the imaginary axis made explicit, the origin of the Euler circle frame is computed twice — under natural coefficients and under imaginary-part (real) coefficients — and both computations give $(0,0)$. The transposition maps the real axis onto the imaginary axis with continuous coefficients and preserves natural coefficients. The frame is fully determined; the analysis of ζ remains the open span of the bridge.

References